

## Short Summaries of Sorted Articles

### Foundations of AI and LLMs (1)

- LLM Techniques and Concepts
  -  schulz-schaeffer\_different\_2025
    - Generative AI differs from other technologies because it is task-agnostic (explicit training with specifics is not needed), requires strategic interaction, and has agency replicating human experiential knowledge.
  -  vargas\_embedding\_2025
    - Dimensionality reduction applied to models shows linguistic structure in data and distinguishes human-generated and AI-generated content.
  -  wang\_finetuning\_2024
    - The superiority of dictionary-based models over BERT in previous work was due to incomplete fine-tuning, and fine-tuned BERT outperforms the alternative.
  -  wang\_instructions\_2022
    - Talks about Super-Natural Instructions, a benchmark of 1,616 NLP tasks, and Tk-Instruct, a model trained on it that outperforms InstructGPT on cross-task generalization despite being an order of magnitude smaller.
  -  wei\_chainofthought\_2023
    - Talks about chain-of-thought prompting, showing that providing intermediate reasoning steps as examples dramatically improves LLM performance on arithmetic, commonsense, and symbolic reasoning tasks.
  -  wei\_finetuned\_2022
    - FLAN demonstrates that instruction-tuning a model on 60+ NLP datasets substantially improves zero-shot (testing samples from classifications not observed in the training set) performance.
- Limitations of LLMs
  -  bender\_parrots\_2021
    - Ever-larger language models carry risks including environmental costs, encoded social biases, and the illusion of meaning in text that is solely pattern-matching.
  -  mccoey\_embers\_2024
    - LLM performance is systematically shaped by their next-word prediction training objective, predicting success and failure based on the probability of tasks, inputs, and outputs.

### Textual and Recording Analysis (2)

- Textual Analysis
  -  barrie\_replication\_2025

- LLM-based research is uniquely difficult to reproduce due to unacceptably high variance in model outputs.
-  bisbee\_gold\_2025
  - As LLMs exceed human expert performance on annotation tasks, researchers should shift focus from chasing state-of-the-art models to assessing the adaptivity of findings to future technological change.
-  burnham\_debate\_2025
  - DEBATE is a family of open-source BERT-based models that match state-of-the-art LLMs on zero-shot and few-shot classification of political texts.
-  chae\_text\_2025
  - Chae evaluates 10 LLMs across four training regimes for stance detection, finding that large models excel in zero-shot settings, while fine-tuning smaller models is cost-competitive.
-  griswold\_staytuned\_2025
  - Uses lexicon-based, supervised, and LLM approaches to perform stance detection on congressional tweets.
-  halterman\_codebook\_2025
  - Uses LLMs against real political science codebooks covering protest, violence, and manifestos to find that zero-shot performance is limited but supervised instruction-tuning significantly improves accuracy.
-  stuhler\_promptbooks\_2025
  - Uses open-source LLMs for structured information extraction from text by converting sociological codebooks into promptbooks.
-  yang\_annotation\_2025
  - Replicates 14 political science studies using 15 different LLMs, finding low intercoder reliability and that LLM choice changes original conclusions in 37% of cases, calling for new annotation practices.
- Image and Video Analysis
  -  crowl\_measuring\_2025
    - Analyzes a decade of local news coverage across 10 U.S. cities, showing that media criticism of police has remained largely stable since 2013.
  -  maranca\_dsl\_2025
    - A design-based supervised learning framework corrects for non-random measurement errors introduced by AI-assisted labeling in image analysis.

### LLMs in Social Networks (3)

- Synthetic Data (Silicon Sampling)
  -  argyle\_simulate\_2023

- Introduces silicon sampling, demonstrating that GPT-3 can accurately simulate the response distributions of diverse human demographic subgroups.
  -  bisbee\_replacements\_2024
    - Tests ChatGPT as a synthetic replacement for human survey respondents, finding that while average scores are plausible, the data lacks sufficient variance and reliability for statistical inference.
  -  broska\_mixedsubjects\_2025
    - Proposes a mixed-subjects design that combines LLM predictions with human subjects' data to preserve validity and reduce research costs.
  -  davidson\_integrating\_2025
    - Generative AI in social science has three core themes: rigorous measurement, optimizing prompting for reproducibility, and novel uses of AI for simulating human behavior.
  -  lyman\_alignment\_2025
    - Compares aligned vs. unaligned versions of six open-source LLMs for silicon sampling, showing that alignment affects outputs in predictable and impactful ways that researchers must account for when choosing models.
  -  weatherby\_silicon\_2026
    - Silicon sampling undermines the purpose of public opinion polling and threatens the integrity of democratic discourse.
  -  xie\_realism\_2026
    - Introduces SSDataBench to evaluate LLM-generated social science data against population-level statistical patterns across seven datasets, finding LLMs compress real-world heterogeneity into oversimplified typologies.
- Social and Cultural Influence of LLMs
  -  arfi\_longitudinal\_2025
    - Combines topological data analysis and generative AI applied to graph neural networks for analyzing longitudinal panel data (repeated observations of the same subjects) in political science.
  -  farrell\_social\_2025
    - Large AI models should be understood as cultural and social technologies with historical parallels to other technologies.
  -  fu\_networks\_2026
    - Uses evolutionary graph theory to show that embedding AI agents in social networks maximizes cooperation, offering a mathematical framework for designing human-AI systems.
  -  salvi\_persuasiveness\_2025

- Finds in a preregistered experiment that GPT-4 is more persuasive than humans in one-on-one conversations.
-  vafa\_wage\_2025
  - Develops fine-tuning methods for foundation models that avoid omitted variable bias in estimation, showing that career history explains more of the gender wage gap than standard econometric models capture.

## AI Bias and Ethics (4)

### ● AI Bias

-  ashwin\_qualitative\_2025
  - Finds that using LLMs to code qualitative interview data introduces non-random bias, and recommends training supervised models on human-coded samples instead.
-  bai\_biased\_2025
  - Develops two psychology-inspired prompt metrics to reveal that explicitly value-aligned LLMs still have biases across race, gender, religion, and health categories.
-  bollen\_biases\_2025
  - GenAI models correlate well with national survey responses when rating neighborhood images but poorly reflect resident sentiment, revealing a systematic bias toward average national perspectives.
-  germani\_bias\_2025
  - Attributing statements to Chinese individuals systematically lowers agreement scores across four leading LLMs, showing that source framing triggers significant nationality-based bias in LLM use.
-  jeon\_demographics\_2026
  - Demonstrates that vision-language models tend to attribute more liberal ideologies to women and Black candidates in campaign ads.
-  mohammadi\_divergence\_2026
  - Compares 17,790 matched Grokipedia and Wikipedia article pairs, finding that Grokipedia leans toward right-leaning religious and historical media sources, so both are biased.
-  tan\_cultural\_2026
  - Investigates cultural biases embedded in AI models and examines how cultural alignment between users and AI affects the quality and diversity of human-AI collaboration.
-  williams-ceci\_attitudes\_2026
  - Shows in two preregistered experiments that biased AI autocomplete suggestions subconsciously shift users' attitudes toward societal issues.

- AI Ethics

-  cheng\_sycophantic\_2026
  - AI sycophancy (being too nice) is prevalent and harmful, reducing users' willingness to take responsibility and fostering over-reliance, yet users still prefer sycophantic AI.
-  costello\_reducing\_2024
  - Shows that personalized three-round conversations with GPT-4 Turbo can reduce belief in conspiracy theories without spreading misinformation.
-  dávila\_racism\_2026
  - Develops a theoretically grounded framework and fine-tuned model for classifying both overt and covert racism in large text corpora.

### Political Applications of AI (5)

- Political Textual Analysis

-  argyle\_testing\_2025
  - Uses AI-generated persuasive messages in two experiments to test theories of political persuasion, finding that all four AI strategies shifted attitudes.
-  benoit\_political\_2025
  - Shows that LLMs offer alternatives to coders for analyzing political texts through natural language understanding, with applications for measuring party positions and policy content.
-  heidejorgensen\_debate\_2025
  - Uses LLMs to analyze online political debates, finding that evidence and a respectful tone improve debate quality, while partisanship consistently reduces it.

- AI's Political Creations

-  argyle\_conversation\_2023
  - AI interventions in online political conversations improve conversation quality and tone without shifting people's policy attitudes.
-  argyle\_persuasion\_2025
  - A review essay synthesizing recent large studies showing that LLMs are highly effective political persuaders, but more persuasive AI tends to produce less factually accurate content.
-  hackenburg\_persuasion\_2025
  - Post-training and prompting techniques boost AI persuasiveness by up to 51%, but more persuasive AI produces less accurate information.
-  waight\_media\_2026

- Government media control shapes LLM outputs via training data, with models generating more pro-government responses in languages of countries with lower press freedom.

### AI's Influence on Labor, Research, Creativity, and Innovation (6)

- AI's Influence on Labor and Research
  -  daniotti\_code\_2026
    - AI coding tool adoption across 160,000+ GitHub developers globally has experienced rapid diffusion and productivity gains for senior developers.
  -  o'grady\_upend\_2026
    - The most sophisticated LLMs can now evade all standard survey quality checks, threatening the viability of online survey research.
  -  westwood\_threat\_2025
    - An autonomous AI synthetic respondent can pass 99.8% of standard survey attention checks while producing plausible and consistent data (manipulation of poll outcomes).
- AI's Influence on Creativity and Innovation
  -  hao\_impact\_2026
    - Analyzes 41 million research papers to show that AI tool adoption gives individual scientists dramatically higher output and citations, but narrows the range of scientific topics studied by 4.63%
  -  sourati\_homogenizing\_2026
    - LLMs are homogenizing human language, perspective, and reasoning by amplifying dominant patterns and marginalizing cognitive styles.
  -  wenger\_creative\_2026
    - LLM responses resemble other LLM responses far more than humans resemble other humans, suggesting that using any LLM for creative purposes risks homogenizing outputs.